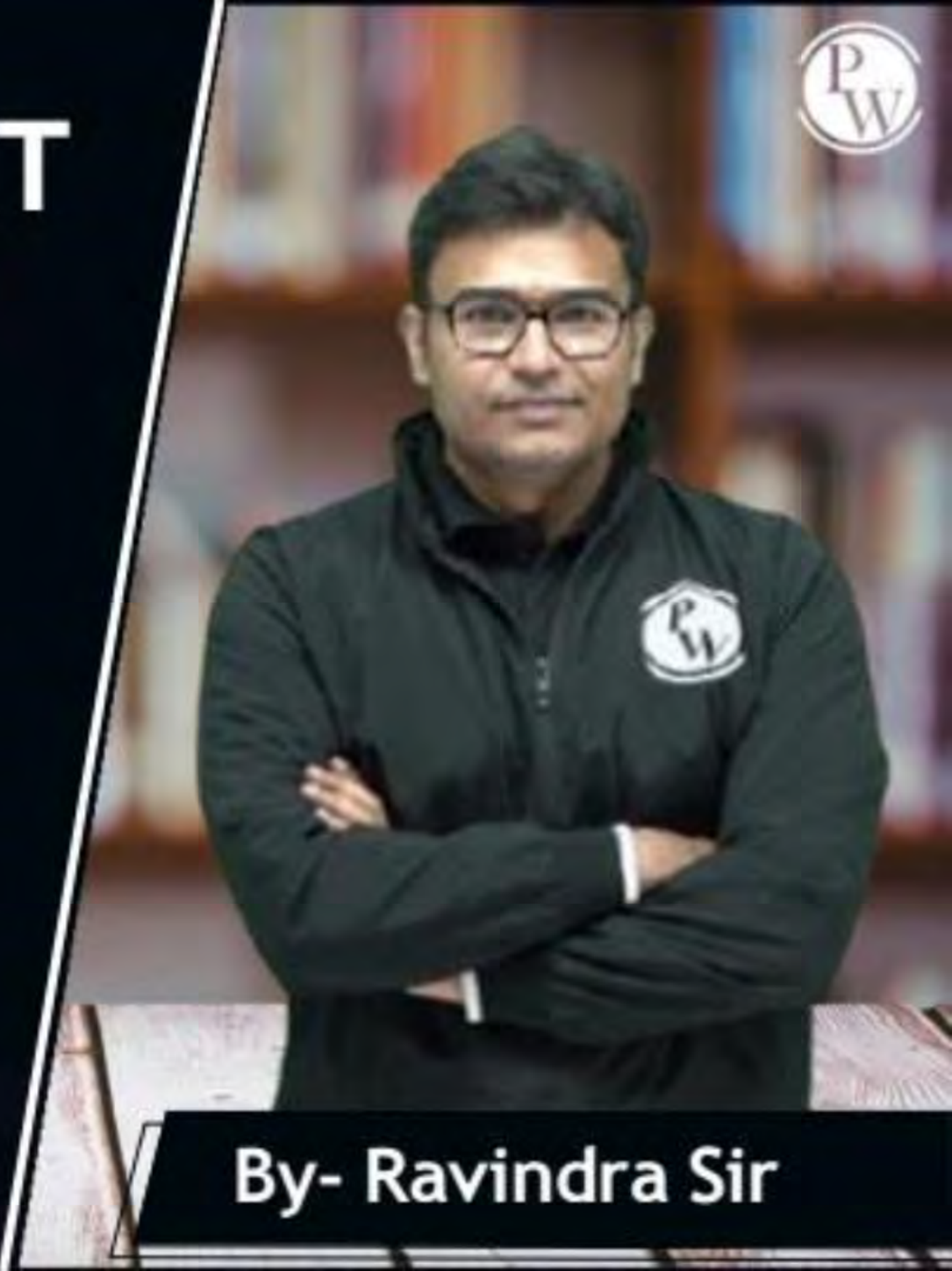
Computer Science & IT

COMPUTER NETWORK
(CN)

PYQ - CN

Lecture No. 01



By- Ravindra Sir

Recap of Previous Lecture



- Topic P X Q
- Topic
- Topic

Topics to be Covered



- Topic
- Topic
- Topic

PYQ

→ In one of the pairs of protocols given below the protocols can use multiple TCP connections b/w the client and the server. which one is that

a) HTTP, FTP
b) HTTP, TELNET
c) FTP, SMTP
d) HTTP, SMTP

→ Identify the correct sequence in which packets are transmitted on the N/W by a host, when a web browser requests a webpage from a remote server

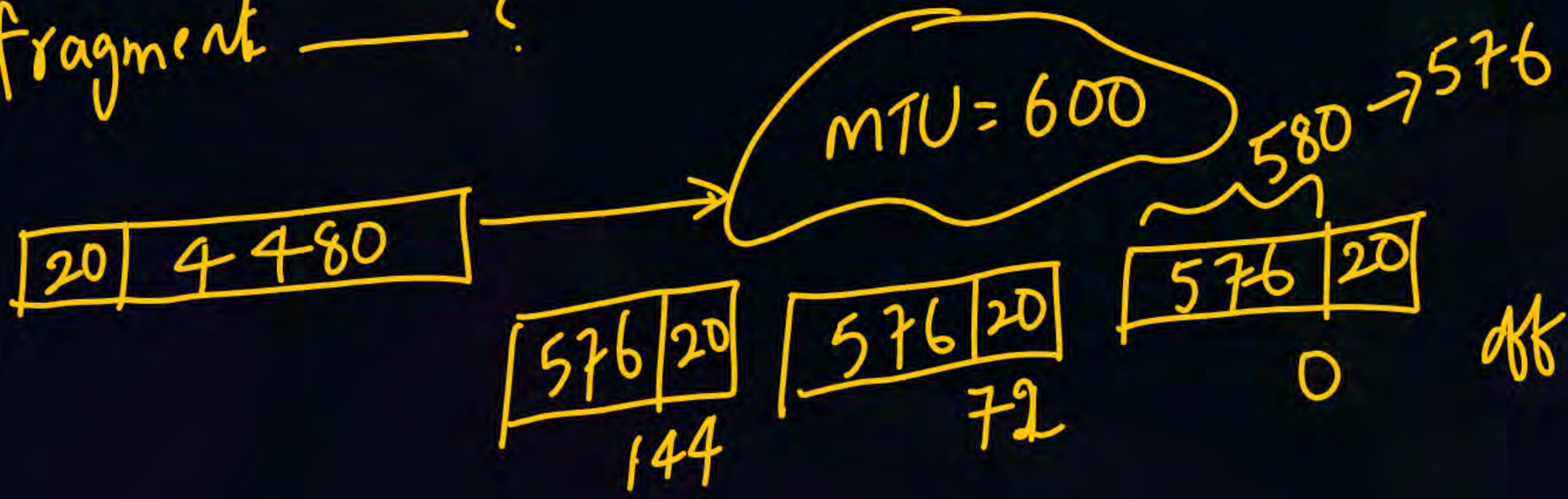
a) HTTP GET, DNS query, TCP SYN

b) DNS query, HTTP GET, TCP SYN

c) DNS query, TCP SYN, HTTP GET

d) TCP SYN, DNS query, HTTP GET

Consider an IP packet with a length of 4500 bytes that include 20 B of IPV4 header and 40 bytes of TCP header. The packet is forwarded to a router that supports MTU of 600 Bytes. Assume fragmentation is done, what is the fragment offset in the III fragment ——?



→ which of the following is true

X a) A router doesn't modify IP packets during forwarding

✓ b) It is not necessary for a router to use any routing algo

X c) A router should reassemble IP fragments if the MTU of the outgoing link is larger than size of incoming IP packet.

d) None of the above

Which of the following pairs can be used to send and retrieve emails (in that order)

a) IMAP, SMTP

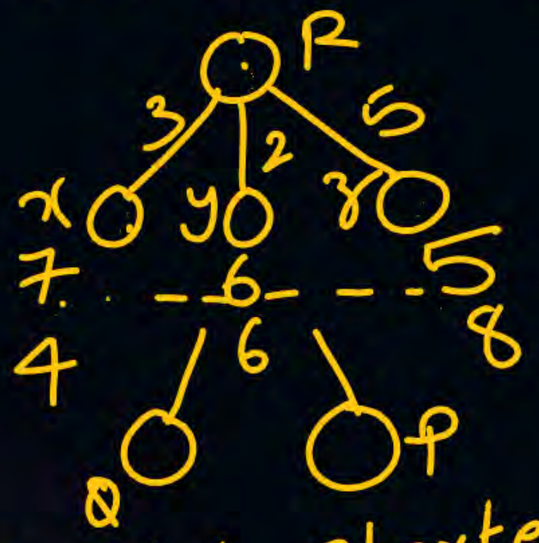
b) SMTP, MIME

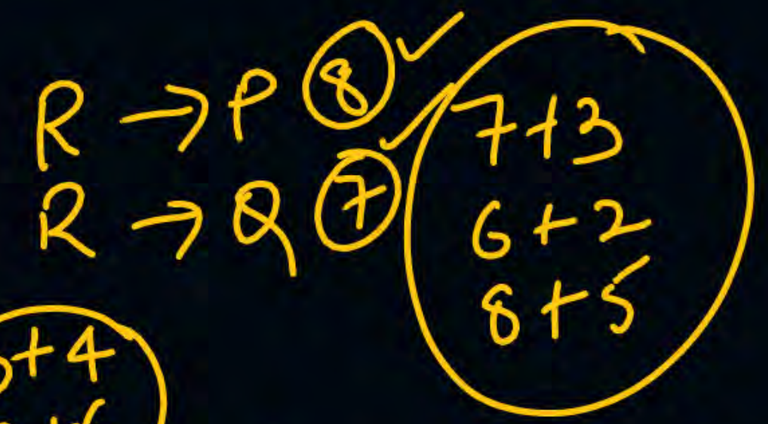
c) IMAP, POP3

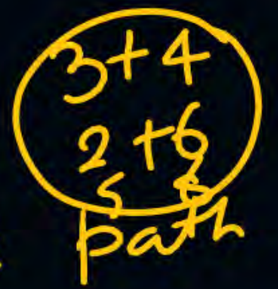
d) SMTP, POP3

Assume that you have made a request for a webpage through your web browser to a web server. Initially the browser cache is empty. The browser is configured to send HTTP requests in a non persistent mode. The webpage contains text and 5 small images. The minimum number of TCP connections required to display the webpage is ____ 6

Consider a Computer network using DVR. The topology is shown below

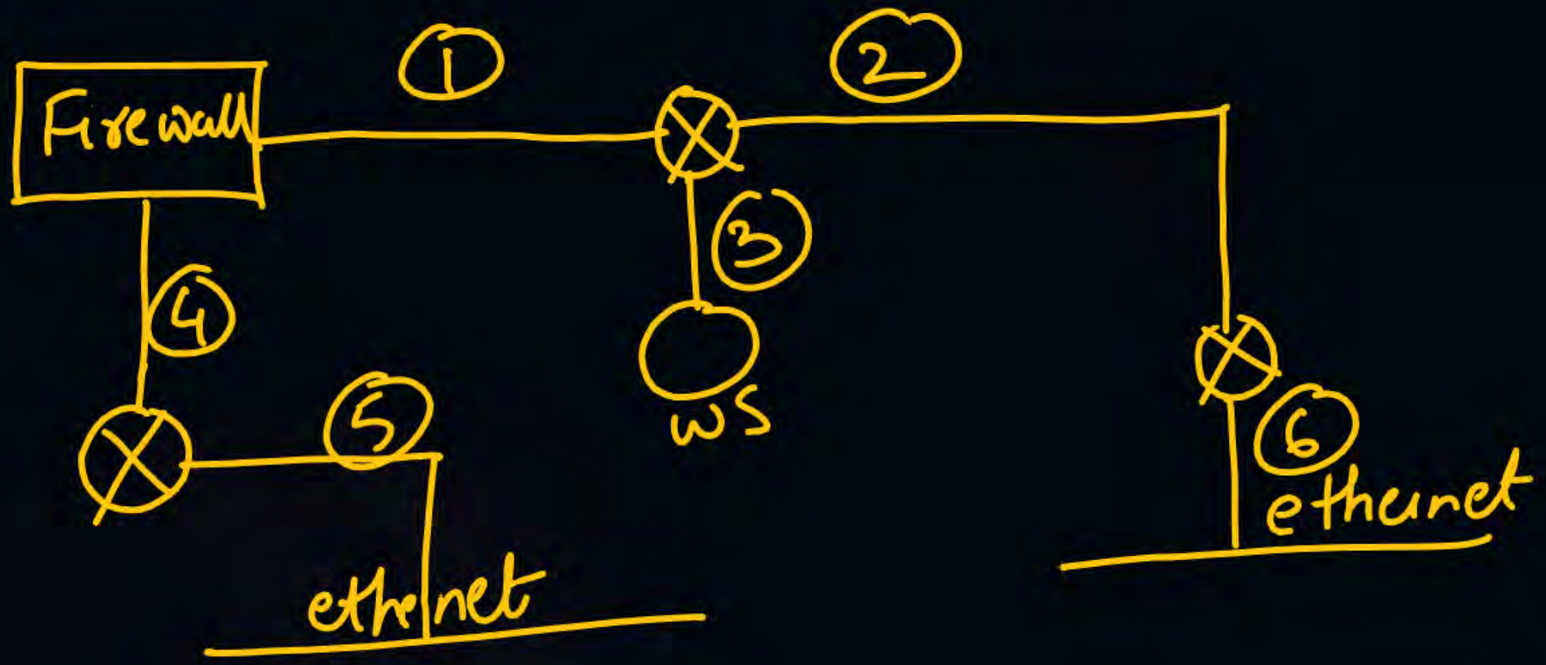






The objective is to find shortest path from R to P and Q

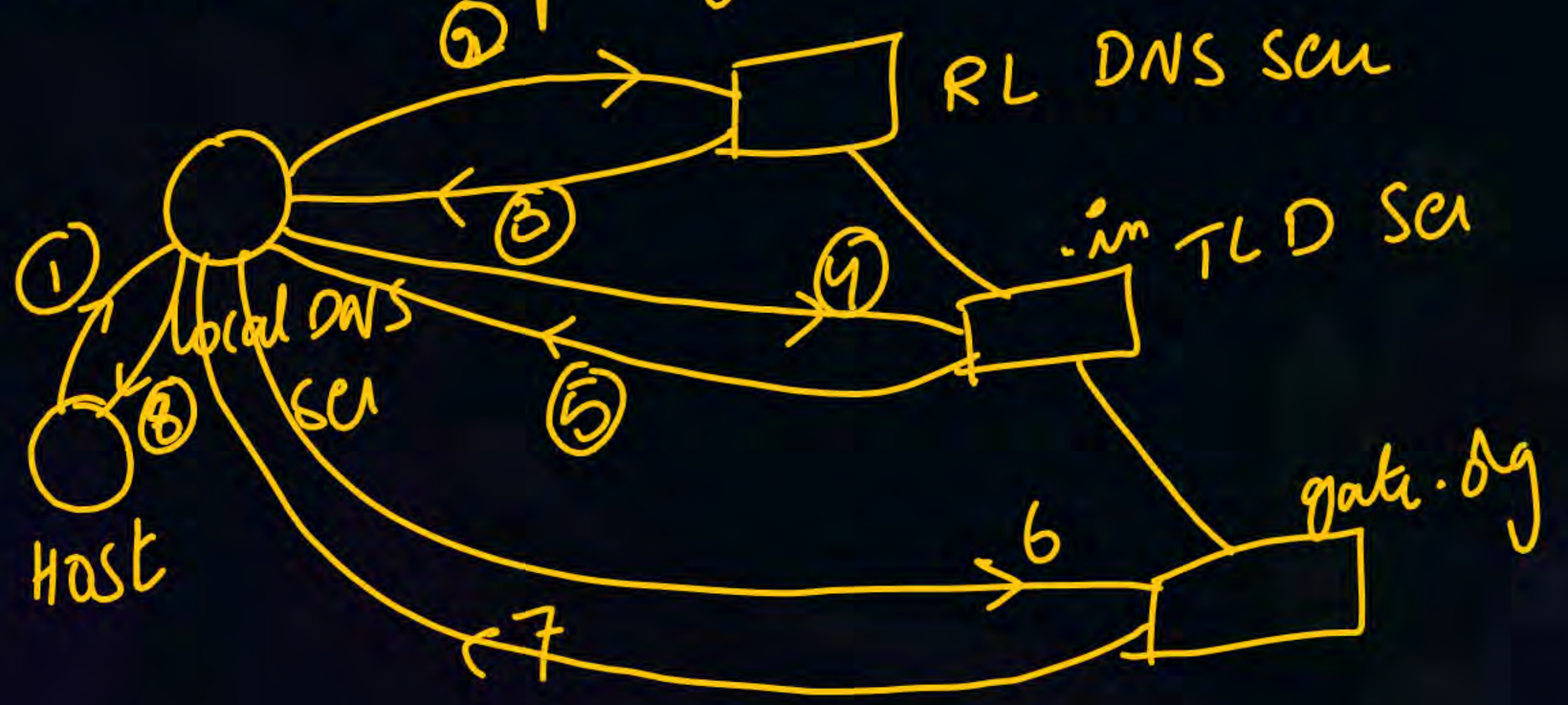
R → X − 3, R − Y − 2, R − Z = 5 Router R gets vectors from X, Y, Z to P as 7, 6, 5 and Q as 4, 6, 8.



no of subnets is ——

a) 3 b) 12 c) 6 d) 8

→ Consider the resolution of the domain name www.gate.dg.in Assume no resource record are cached and iterative query mechanism is used. The number of DNS query-response pairs involved in completely resolving the domain name



→ A user browsing a webpage hosted at a remote server. The browser opens a single connection to fetch the entire webpage. The webpage consists of top level index page with multiple image objects. All caches are empty. The following packets leave the user's computer in which order

(i) HTTP GET for index page

(ii) DNS query

(iii) HTTP GET request for an image object

(iv) TCP SYN to open connection

a) iv, ii, iii, i b) iv, ii, i, iii c) ii, iv, i, iii d) ii, iv, iii, i

Consider the routing protocols in List-I and the names given in List-II

List I
(i) DVR
(ii) LSR

List II
a) Bellman ford
b) Dijkstra

a) (i)-a and (ii)-b
b) (i)-a and (ii)-a
c) (i)-b and (ii)-a
d) (i)-b and (ii)-b

Choose the best matching b/w list (i) and list (ii)

List I	List II
P: DLL	1. ensures reliable transport of data over physical point to point link
Q. NL	2. Encodes/decodes data physical transmission
R: TL	3. Allows end to end communication b/w two processes
	4. Routes data from one n/w to the next

	P	Q	R
✓a)	1	4	3
b)	2	4	1
c)	2	3	1
d)	1	3	2

→ In a DLL protocol, the end frame delimiter flag is given by 0111. Assuming that bit stuffing is employed the transmitter sends the data sequence 01110110 as

a) 01101011 b) 01101011 0

c) 011101100 d) 011010100

01101011 0

0111 ✓

011 '0'

011 0

THANK - YOU